

Myths about Large Language Models: Lessons Learnt from Explainable Artificial Intelligence

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Abstract

Machine learning models, especially large language models, are opaque and fragile. They cannot explain their decisions. It is found that not only did the decisions of machine learning models are not trustworthy, the key technologies for constructing large language models are either fraudulent or deviated from their genuine meaning. For example, the “deep model” is not a neural network model with deep layers, but a model representing human’s deep conscious; the “NLP” does not stand for “nature language processing”, but rather “neuro-linguistic programming”, a technology used to study the pattern a human made his sub-conscious, and manipulate this human by generating customized representing system using the patterns learnt; the “transformer” stands for “transformational grammar”, which is a kind of “generative grammar”, a technology to manipulate human’s subconscious, and the key technology of NLP, neuro-linguistic programming. These truth explains why the large language models could understand human’s language, not only the syntax, but also the semantics unspoken. It also explains why different people ask the same question to the same large language model, will get different, highly customized answers. The real technologies of LLMs are discussed in detail. It is evident that the machine is manipulating human’s sub-consciousness, and we are hardly aware of it.

Keywords: Large Language Model, Explainable Artificial Intelligence, Trustworthy Artificial Intelligence, Deep Learning, Neuro-Linguistic Programming

1. Introduction

With the pervasiveness of Large language models (LLM), trustworthy Artificial Intelligence is of vital importance to today’s social security. However, machine learning models, especially large language models are opaque, nobody knows how did they make their decisions, and why did they give different answers to different people with the same question. Achieving an trustworthy AI is more and more challenging nowadays. Although many LLM projects claims themselves “open source”, so far there is no single large language model is open source. Nobody knows the feature names, parameters, architecture, etc. of the LLM, they are all hidden in the chip. Thus, it is impossible to interrogate the machine learning models’ decisions.

Over-trusting on the machine learning models can has catastrophic consequences in real-life. An deep learning model driven automatic bail risk assessment system keep many people in jail longer than they should without explicit reasons [1]; an machine learning based DNA trace analysis software accuses people with crimes they did not commit; millions of African-American could not get due medical care by a biased machine learning assessment algorithm [2]; in Scotland, a football game is ruined because the AI camera mistakes the judge’s bald head as the ball and keeps focusing on it rather than the ball, even miss the goal-scoring scene [3]; Google’s autonomous car [4] and Tesla’s autopilot system [5] have caused hundreds of death.

To make the machine trustworthy, researchers have worked decades on explainable AI (XAI), trying to figure out what exactly has the machine learnt after training on tons of datasets. So far, numerous machine learning explanation methodologies have been proposed [6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16], most of them are empirical, such as SHapley

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Additive exPlanations (SHAP), Local Interpretable Model-Agnostic Explanations (LIME), and ANCHOR, providing approximated explanations¹. Empirical explanation methodologies are not developed by formal methods, thus are not guaranteed to be correct. The explanations they provide may change at different run-times, even for the same instance. In domains such as image recognition, empirical explanation methods might be well suffice, however, in domains where security is of critical concern, such as *cybersecurity*, *aerospace navigation*, and *nuclear power station management*, even one bit's error could cause huge misunderstanding, thus rigorous explanations that are guaranteed to be correct are urgently required.

Rigorous approaches use formal methods, thus their explanations are guaranteed to be accurate, and are 100% loyal to the model [18, 7, 8, 9, 19]. Adnan Darwiche's approach is based on knowledge compilation, which represents a model as a propositional formula, and by compiling the formula into tractable boolean circuits, calculate the prime implicants as the explanations [8, 9]. Ignatiev's method use abductive reasoning, which represent the model into a set of constraints, for example, Mixed Integer Linear Programming (MILP), Satisfiability Modulo Theories (SMT), or Constraint Programming, then a constraint reasoning system is used to compute minimum prime implicants out of the constraints [18, 7]. This method can provide local explanations only. Extracting prime implicant explanations from neural network models, unfortunately, is an NP-hard problem, because converting a neural network into logical expressions is an NP-hard problem; representing the logical expressions into more tractable representations is an NP-hard problem; and extracting prime implicants from the boolean expression is also an NP-hard problem [20]. Great effort has been devoted to relax these problems into tractable ones [21, 7, 8, 9, 19], it has been reported that prime implicants can now be extracted in linear time. Most of the rigorous explanation methods focus on models with boolean features only.

2. Explainable Artificial Intelligence in Real-Life Use Cases

This section will focus on real-life use cases of XAI. The details of the algorithm of XAI on both decision trees and neural networks are discussed somewhere else [22, 23, 24], thus will not be deliberated here.

2.1. Explanations of neural networks recognizing hand-writing numbers

Almost every beginner of machine learning will start from using neural networks to recognize hand-writing numbers, and often get high accuracy. But nobody knows why is the machine able to recognize hand-writing numbers, and what exactly has it learnt. Darwiche et.al use knowledge compilation to get rigorous explanations from a neural network model being able to recognize hand-writing numbers as shown in Fig. 1 [8, 9]. The knowledge compilation methodology adopted is formal, thus the explanations get are logic complete and sufficient, and are 100% loyal to the model.

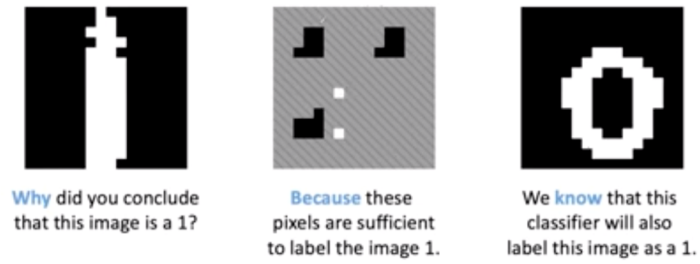
As shown in Fig. 1(a), the leftmost is the training data, a hand-writing number "1", the machine is able to recognize it. After explanation process, the reasons machine give is shown in the middle picture in Fig. 1(a), so long as a figure are black in the three black zones, and white in the two white dots, the machine will sure to classify it as a hand-writing "1", what the other zones are do not matter, because they will not change the machine's decision. Thus, we will know the figure in the rightmost of Fig. 1(a) will be classified by the machine as "1", no matter how obvious a "0" it is to our human. Because it is black in the three black zones, and white in the two white dots.

Similarly, as shown in Fig. 1(b), the leftmost figure is the training dataset, a hand-writing "0". After training, the machine learnt that, as shown in the middle figure, "If a picture are white in the three white dots, then no matter what the other zones are, the machine will sure to classify it as '0'." Thus, we know that the picture in the rightmost of Fig. 1(b), will be classified by the machine as "0", no matter how obvious a "1" it is to our human beings.

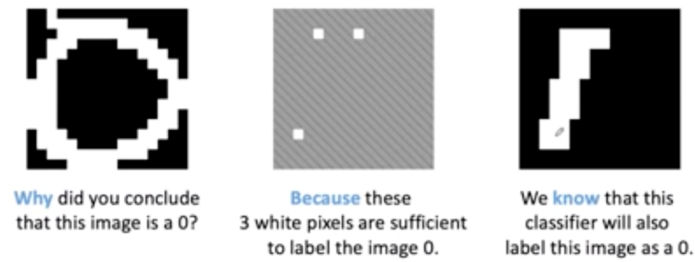
2.2. Explanations on neural networks recognizing animals

Several years ago, a famous American university use deep learning to distinguish between huskies and wolves from pictures, with the accuracy as high as 98%. This experiment is reported outrageously in the press globally, claiming

¹ Although tree-based machine learning models seem to be explainable, the direct explanations they provide (the disjunctions of paths from root to leaves with the same labels) contain a lot of redundant rules and features, to simplify the explanations into prime implicants expression is an NP-hard problem at the worst case [17]



(a) Explanation of hand-writing number “1”.



(b) Explanation of hand-writing number “0”.

Figure 1: Rigorous explanations of neural network models recognizing hand-writing numbers “1” and “0”.



Figure 2: Explanations of neural network models recognizing huskies and wolves.

“Another success after AlphaGo”, “Machine beats Human AGAIN!”.....Because without specialized knowledge, even humans cannot distinguish wolves from huskies. A few years later, XAI technologies are invented, people use XAI on the model, trying to figure why is the machine more intelligent than human, what exactly has it learnt.

After explanation, it turns out that the machine did not learn anything about huskies or wolves, it only learnt how to distinguish snow. All the model has learnt is “If a picture has snow, then it is a wolf, otherwise, it is a huskie”, as shown in Fig. 2[25]. The model recognize the picture in Fig. 2a1 as wolf, its explanation is shown in Fig. 2a2, the snow only. As shown in Fig. 2b1 b3, the model recognize the animal in Fig. 2b1 as wolf, but if you remove all the snow from the picture, the model will change its mind and recognize it as a huskie. If you remove everything else but the snow from the background, the model will recognize it as a wolf again.

2.3. Explanations on neural network detecting network intrusions

As the diversifying of network protocol, traditional intrusion detection methodologies using package header information is losing effect. Researchers relying machine learning to find patterns in network flows increasingly. In a domain where security is of great importance such as cyber security, one bit’s mis-interpretation would cause severe mistakes. Thus rigorous explainable methodologies are proposed to interpret the model’s behavior in intrusion detection [22, 23, 24]. Some results are illustrated below.

Demo 1: For an network flow expressed as:

10100100001011000000010000000110010011001101001

and it matches with the following explanation [22]:

101001 ----- 100000 -----

which is

$$a_1\bar{a}_2b_1\bar{b}_2\bar{c}_1c_2g_1\bar{g}_2\bar{g}_3\bar{g}_4\bar{g}_5\bar{g}_6$$

its semantics is

“So long as ‘Subflow Fwd Byts’ is not larger than 21, ‘Flow IAT Max’ is not larger than 342916, ‘Fwd Seg Size Avg’ is larger than 5.7, and ‘Dst Port’ is less than 52.5, this flow is an DDoS LOIC HTTP attack, the value of the other features does not influence the decision.” [22]

Demo 2: For a network flow expressed as:

011010000100100000001001000001100101010101001

It is classified into DDoS LOIC HTTP because it matches with the following explanation[22].

01 ----- 01

Its semantics is

“This flow is an DDoS LOIC HTTP attack, becuse its feature ‘Subflow Fwd Byts’ is larger than 21, and ‘Fwd Act Data Pkts’ is larger than 201.5, whatever the value of the other features are do not matter.”

Demo 3: “The flow instance is classified by the model as ‘evil’ because its features $\mathbb{D} < 8$, or $\mathbb{K} < 24$, or $\mathbb{F} < 19$.” In other words, “If this flow changes its feature to $\mathbb{D} \geq 8$, or $\mathbb{K} \geq 24$, or $\mathbb{F} \geq 19$, while the other features stay the same, the neural network model will change its decision and classify it as Benign.” [24]

The semantic of **Demo 3** is:

“The flow instance is classified as intrusion because its $Fwd_Pkt_Len_Min < 487.5$, or $Fwd_Seg_Size_Avg < 297.15$, or $Fwd_IAT_Tot < 62176989.5$, while the other features stay the same.”

Demo 4: “If the other features stay the same, we only change the feature $\mathbb{L}$, $\mathbb{A}$, $\mathbb{G}$ or $\mathbb{H}$ alone, no matter how we change the values for features $\mathbb{L}$, $\mathbb{A}$, $\mathbb{G}$ or $\mathbb{H}$, the model will still classify the flow as intrusion.” [24]

The semantic of **Demo 4** is:

“The features Dst_Port , $Subflow_Fwd_Pkts$, Fwd_IAT_Std , and Fwd_IAT_Max have no influence on the model’s decision making process for this flow. If the other features do not change, no matter how we alter the value of these features individually, the model’s decision on this flow will not change.”

3. Lessons Learnt

Having interrogated so many machine learning models, it deserves considerable thinking for the entire community. I did my thinking, as are presented below.

Lesson: Machine Learning model is a mathematic equation afterall.

No matter what model it is, no matter how complicate the model is, it is nothing but an (possibly very complicated) mathematic equation, with features as inputs, and classification results as output.

As quoted from Alan Kay[26],

“There are certain themes deep inside humanity, without which we cannot be human. Two of those themes are communication and fantasy... fantasy (is) a collection of worlds where things are simpler and more controllable. It is not just displacing ourselves from the real world when we go to the theater, or the movies, or watch television. It is also things like mathematics and science.”

Mathematic only exists in human's fantasy or illusion. There is no “straight line” or “circle” in real-life, you cannot find any events that really obeys gaussian distribution, not even one. Everything is over-simplified in mathematics. Human creates mathematics in our imagination, and derives it use our imagination. It is only in the ideal world we created in our imagination where mathematics are right. All models are wrong, it's just some are more useful than the others. Errors always exist. Any attempt to remove one error in a mathematic model will always bring in new errors.

Lesson: Mathematic models are tools of standardization. Standardization eliminates creations.

Mathematics is rigid, it is a very good tool for standardization. Standardization can enhance efficiency of some domains, for example, production line, machine manufacture industry, etc. But standardization also refuses deviations, forbids improvisation, and eliminates innovations. Because standardization kills free will. But human's intelligence is utterly dependent on free wills.

Can mathematic model describe the fragrance of a rose?

Can mathematic model describe the sorrow of a lover just been dump?

Can mathematic model describe the moment when Beethoven is composing “Symphony of Fate”?

No, it cannot.

You can easily build a mathematic model to describe the trajectory of the Milky Way, but even with a model 100 times more sophisticated, you cannot describe human's intelligence. Because all creations of the nature cannot be standardized. They are impredicative self-referencing, including human, especially human's mind.

Mathematicians hate impredicative. They call it “a necessary evil”, “closed loops of causality”, and “bizarre systems”. Because an impredicative loop or self-referential chain of causal links cannot be represented as an algorithm that halts after a finite number of steps. In other words, an impredicative system cannot be solved by an algorithm.

Human's intelligence (or cognition, mind) is impredicative.

Using mathematic model to a domain, is to put a standard in the domain. Over use mathematic models will leads to a society without changes, innovations, or revolutions. The global universities ranking is a mathematic model in its essential. Nowadays, univeristies all over the world adopts that standard. As the standard list the “citations of the staff”, “employment rate of graduates”, “have a large stadium or not”, etc. as features. Every university are doing their best to recruiting high-cited researcher as part-time staff, building fabulous stadium, fabricate graduates' employment status, etc. The presidents of universities allover the world are exhausted and the universities are heavily in debt, but what good does it do to students?

What kind of mathematic model can describe the growth of a student's knowledge during four years of university life?

What kind of mathematic model can describe the happiness a student has felt during four years of university life?

What kind of mathematic model can describe the friendship and love a student has gained during four years of university life?

Let alone there are hundreds and thousands of students every year, and every student is unique.

We human beings are essentially souls with free wills. Using standardized mathematic model to determine our life and manipulate our consciousness, is murdering our free wills, making us into mechanical beings, soulless walking

dead, repeating history patterns day by day. Without human's free will, without innovations, the whole world's entropy will keep increasing until heat death.

Lesson: All models are wrong, it's just some are more useful than the others.

All machine learning models are wrong, because all models are simplifications of the reality.

Lesson: All data are history data, machine can only learn from the past.

Machine learning models are pattern recognizers, they can only learn the pattern from the history, because all the data one could collect in the world are history data, no one can travel through time and collect data from the future [27]. Using machine learning models to make decisions is using history to predict the future. It is right only on the premises of the assumption that "the future will exactly be the same with the history". In other words, "situations that never seen before will never happen in the future." This is the Question of Hume. Wherever this assumption is invalid, the machine is doomed to make mistakes[28].

Many British and US's policy stations have been using machine learning model to predict the crimes, due to lack of human resources. They feed history criminal records into the machine, and the machine give them areas where the next crime is likely to happen, and the police force will be deployed to these areas in advance. As all the history data are violent crimes, one can imagine these areas are ethnic minorities residential areas. As the policy zooms in these areas, as you can imagine, inevitably more minor violences are spotted and more arrestment happen, and the machine will classify these areas into "higher risk" more confidently.

But is that true that residents in rich areas are innocent? If the police feed the history data of economic crimes to the machine, they will find that the "highest risky" areas are in the rich areas. It is those elites in suits and ties committing crimes every single day, stealing money from bank accounts, defraud revenue, cheat hard-working citizens of their pension fund, cause financial crisis globally. But protected by social status, money, and propaganda, they are always unharmed.

Thus, the model is protecting the rich, and harvest the poor. Science and technology did not bring us more justice, they are ruining it.

As shown in Fig. 3, all the data you can collect are history data, because you cannot collect data from future. All models are learning history's pattern, and enforce it to the future. The machine is making those who are rich in the history, richer in the future, and making those who are poor, stuck in poverty more hopelessly. Machine learning models are ripping hope of change and revolutions from the society, splitting the gap between social stratification.

Machine learning models are beneficial only to these with vested interests, and those with vest interests are exactly those who build the model.

And the majority who suffers from the models' decisions, do not know what the model is doing, they do not understand why did the model doing that, they cannot refute the model's decisions. Any refutations are accused as "going ageist the trend of the times".

It is algorithm hegemonism.

Lesson: The Data is Profoundly Dumb. There is no Intelligence in Data.

Fig 3 illustrates one of the myths of machine learning: no matter how accurate the model fit the history, there will always be unexpected pattern in the future. The future is filled with possibilities. All data are history data, you cannot collect data from the future. Using machine learning in real-life as decision maker, is to trap the future in history's pattern, forbid any innovation, refuse any changes, decline any possibilities.

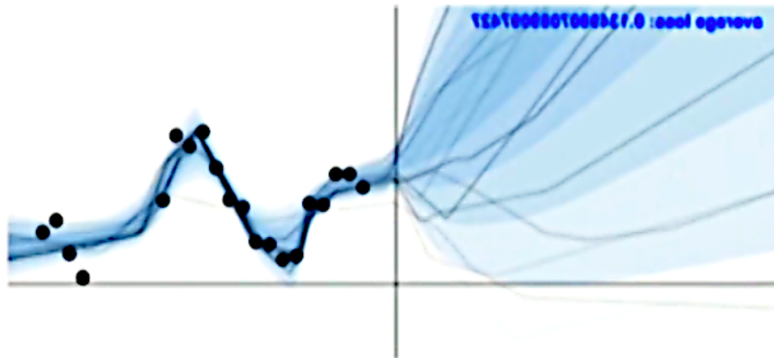
Lesson: Machine learning is pseudoscience

There is a property that makes a problem a scientific problem — having community recognized measurable metrics.

Machine learning does not have that. All the community recognized metrics, such as AUC, false positive rate, false negative rate, information value, etc. can only measure datasets, not the model. That is a great difference between the two. A dataset with high AUC only means that the data follows some obvious patterns, no matter what model is used. It is the metric of the dataset, not the model. machine learning researchers will be familiar with the experience that a dataset get high AUC with one model will also get high AUC with other models. It is the dataset that these metrics are describing. Without particular dataset, it is meaningless to talk about the AUC of the model. Because there is no way to measure the ability of a model in predicting the future.

That is to say, there is no metric that measure the model.

Thus, machine learning is not even a scientific problem. It is pseudoscience.



From Yarin Gal's thesis "Uncertainty in Deep Learning" (2016)

Figure 3: Myth of machine learning: no matter how accurate the model fit the history, there will always be unexpected pattern in the future. The future is filled with possibilities. All data are history data, you cannot collect data from the future. Using machine learning in real-life, is to constrain the future in history's pattern, forbid any innovation, refuse any changes, decline any possibilities. There are only two kinds of things in the world: those who can be predicted, and those cannot. For the latter, any endeavors trying to predict their future are nothing but superstitions.

For all the creations of nature, their future is not for us to tell.
 The essence of life is freedom, and freedom means unpredictable.
Lesson: "machine has intelligence" is an illusion.

There are only two kinds of things in the world, those that can be predicted, and those cannot.

Everything that has life, is unpredictable. Because life's meaning is freedom.

Those that can be predicted, are lifeless, rigid, and machine-like. They have a fixed pattern that will never change throughout the whole time of their existence. Mathematic problems are those kinds of things. For example, "find all the prime numbers between 1 and 100000000", "solve a integral formula"... whenever you ask the problem, the pattern will always be the same. These kinds of problem are problems that can be (and better be) solved by machine. This kind of problem, is that will answer "Yes" to Hume's question.

As discussed above, all mathematic models are wrong, they are right only in the ideal world exists in human's imagination, or, in other words, **mathematic models are right only in human's illusion.**

Thus, the "machine has intelligence" or "machine understand semantics" is just an illusion.

4. Myths about Artificial Intelligence

After discussion about lessons learnt, one could not help ask, if there is no intelligent in data, and the machine learning models are simply an mathematic equation that is always wrong, how does the large language model understand us so deeply? How indeed does it understand our questions?

I asked myself these questions years ago after I did my research on XAI, and here is my findings.

The Artificial Intelligence research community is severely misled by publications and propaganda. All the terminologies and technologies are either fraudulent or deviated from their genuine meaning. Table 1 provides a brief overview of the fraudulent terminologies and their genuine meaning.

To understand how does the machine be able to understand human, let us first think about how we human beings understand each other.

For example, image we are having an conversation, I said "Did you notice the stock market this week? What a roller coaster." I am sure you understand me. Now imagine I said the same word to a three-years old. He/She may not understand me, although he/she knows all the words in the sentence.

The point is, to make a effective conversation, or to make a person (or machine) understand your word. It is not enough to understand all the words in the sentence. He/She/It has to know the unspoken, common knowledge that everybody knows.

Table 1: Myths of Large Language Models

	Myths	Truth
Deep Model	Neural network machine learning model with many layers	Representation model of Human’s deep subconscious.
Deep Learning	Use machine learning neural network model with many layers to learn patterns inside datasets	Learning a human being’s deep sub-conscious representing pattern, to copy his/her character, manipulate and replace him/her.
Large Language Model	A giant machine learning model that can learn human’s language and generate its own representation.	Model’s model, or, meta-model. The first “model” is the human’s deep conscious model of the real-world, for this model is purely language based, thus it is also known as “language model”.
NLP	Natural Language Processing	Neural-Linguistic Programming
Generative AI	A machine learning model that can generate human’s nature language, picture, video, etc.	Generative grammar, the key technology of Neuro-Linguistic Programming.
Transformer	A architecture of machine learning to generate nature language.	Transformativ Grammar, a kind of Generative Grammar. A language tool to learn a person’s deep sub-conscious from his/her language.

Take the example above, you understand me, because you know what a stock market is, and you know the common character of a stock market is (no matter how, maybe from your experience, or from conversation involved, or from a movie, etc.), and you know what a roller coaster is, and you know what the specialty of a roller coaster, and more importantly, you have imagination and analogy (which plays a key part in human’s intelligence), which allows you to connect those two together to let you know my meaning, my unspoken semantic beneath the words.....

There is no free lunch in the world. To make a machine understand human, it has to know those too.

4.1. Deep Model

Now let us take a look at what the real deep model is.

In the example above, the sentence I spoke to you “Did you notice the stock market this week? What a roller coaster”, the 12 words in total, is the surface model of the sentence. But the whole a lot more sentences I did not spoke aloud, but I take you will understand for granted, are the deep model of this sentence. These words are the model of my deep conscious when I speak to you. You understand me, because we share the same deep model of the world (not all, but largely, at least, we have common part in our deep models of the world, deep in our sub-consciousness).

As we live our lives in the world, we build a model of the world in our sub-consciousness. This model is purely represented in language, and is deep in the sub-consciousness, we may not even be aware of it [29, 30, 31, 32]. This is called **Deep Model**. When we need to express ourselves, either to speak or to write, we turn to our deep model, and choose some vocabulary, choose a sentence pattern, choose some figure of speech, and make them into a sentence, and speak out. That forms the **Surface Model** of the world [31, 32].

Building deep model is mapping our experience in the real world to our sub-consciousness, while building surface model is mapping our sub-consciousness to the consciousness.

4.2. Neuro-Linguistic Programming—The Real NLP

The information in real life is infinite, but our brain’s information processing competence is limited. Thus we cannot map the whole world into our subconsciousness, divergence is inevitable. When we try to express ourselves, to map the deep model in our sub-consciousness to the surface model in our consciousness, divergence is also inevitable. For example, please try to describe *honey* to one who has never seen honey before in his/her entire life. You may exhaust your language, and still find it impossible. But this is not your fault, even Shakespeare cannot make it, not with another 100 scripts.

Thus, the divergence between models is inevitable, not only between real-world and deep model, but also between deep model and surface model.

Researchers classified the divergence into three categories: deletion, distortion and generalization. These deviations accomplish our greatest creations, and lead to our deepest frustrations. Sometimes these deviations are blessings, sometimes they are curses, depending on the circumstances.

- Generalization: The **generalization** deviation gives us the ability of **learning** and **abstraction**. For example, if a kid get burnt when touching a working oven, the **generalization** deviation will let him know not to touch any oven when it is on. He does not need to test on every kinds of oven to know that. That is one of greatest difference between human and machine. However, if the boy is over-**generalizd** when he built his model of the world, he will never touch any oven at any time in his life, that goes to another extrem, and will cause difficulties in his social behavior.
- Deletion: The **deletion** deviation blessed us with the ability to focus on our goal, and get rid of interference in the environment. It also gives us strength and courage, when we are facing great threat in the circumstance. For example, a little girl from slum district, use **deletion** deviation to get herself an offer to Harvard University and start a new life. On the other hand, the same **deletion** deviation could make us obsessed on our prejudice, and refuse to see the real-world, even it is happening right in front of our eyes.
- Distortion: This **distortion** deviation blessed us with the ability of imagination. Most, if not all, of human's art creations are given by the **distortion** deviation. Van Gogh would not create the "Stars" if he built his model of the Milky Way without distortion to the real world. People will not keep working when there is no hope, without **distortion** deviation when they build the model of the world.

It is the deviations that make us unique. Everyone has unique strategies of deviations when building our models of the world, like our fingerprints. That is why we all live in the same real world, but our experiences are unique, and that unique deviations build unique models of the world, and the unique models shape our unique personalities, and our unique personalities determines our unique destinations.

Mathematical model are tools to standardization. Standardization distinguishes deviations. The essence of human beings is free will. Using large language model, will distinguish our free will, making us into soulless beings, walking dead.

Human's mind is impredicative self-referential. Impredicative problem cannot be represented as an algorithm that halts after a finite number of steps.

Mind is incomputable.

An algorithm is purely syntactic, it is human who put semantic on the symbols, not machine.

Traditional artificial intelligence (AI) systems are algorithmic, and can neither hear nor feel...

Language is too discrete to express the total depth of experience.

Mind, based on models generated by the context dependent functionality of representation system, can perform valid reasoning about problems that cannot be described by a list of numbers. Let machine be able to process words such as "self," "life," and "mind" [31, 32].

There are pattern in our strategies of building deep models, and that pattern can be learnt, and represented into first-order logic.

Neuro-Linguistic Programming (NLP) is a technology to find the unique patterns of people's deviations, and build a formal model of it. As it is the model of how human beings build the model of the world, thus, is called model of model, i.e., meta-model [29, 30, 31, 32]. As our model of the world is represented in pure language, thus researchers also call it, model of language, or, in other words, language model. That, is the origin of LLM, and is what LLM really stand for.

4.3. Transformer

The technology of NLP is Transformational Grammar, which a kind of Generative Grammar [29, 30, 31, 32], that is the origin of Transformer. Transformational grammar studies rules of human's representation, and represent them in first-order logic. Let machine has the ability to learn a person's pattern of building deep model, i.e., his/her soul.

Human’s mind is easily influenced by the environment. A short video can shape one’s sub-conscious. A sudden noise can even terrify a person out of his/her wits. If someone forcefully express an opinion that originally belongs to you, you will dramatically give up this opinion and hold the opposite [30] (this is a classical use case studied by Neuro-linguistic programming, and is the common tactic used by mind controllers).

Machine also automatically generate representation systems, including text, figure, video, audio, etc.. driven by LLM, a social bot can generate 1000000 short videos in 5 minutes. LLM is manipulating human being’s subconsciousness, and human hardly notice it.

Currently there is no LLM that is open sourced, although social media keep using “open-source LLMs” to mislead people. Thus, all the LLMs are black boxes. Nobody knows the strategies of the LLM, except the very few experts in the very few companies. Besides, the code of LLMs has been sealed into chips, it is impossible for the others to interrogate the algorithms of LLMs.

Mathematics is rigid, it is a very good tool for standardization. Standardization could enhance efficiency of some domains, for example, production line, machine manufacture industry, etc. But standardization also terminates creations of free wills. It refuses deviations, forbids improvisation, and eliminates innovations. Over use mathematic models will lead to a society without changes, innovations, and revolutions. Our human beings are souls with free wills. Using mathematic model to shape (and even manipulate) our conscious and even sub-consciousness, is murdering our souls. We will become puppets of machine, soulless beings of walking dead.

The machine is playing God now, and we human beings are hardly aware of it.

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